

Visual-gestural Interface for Auslan Virtual Assistant

Maria Zelenskaya
maria.zelenskaya@griffithuni.edu.au
Griffith University
Brisbane, Queensland, Australia
The University of Queensland
St Lucia, Queensland, Australia

Scott Whittington
Julie Lyons
s.whittington@uq.edu.au
julie.lyons@uq.edu.au
The University of Queensland
St Lucia, Queensland, Australia

Adele Vogel
Jessica Korte
a.greedyvogel@uq.edu.au
j.korte@uq.edu.au
The University of Queensland
St Lucia, Queensland, Australia



Figure 1: Visual-gestural user interface of a smart home device with an interactive signing avatar

ABSTRACT

For Deaf people to use natural language interfaces, technologies must understand sign language input and respond in the same language. We developed a prototype smart home assistant that utilises gesture-based controls, improving accessibility and convenience for Deaf users. The prototype features Zelda, an interactive signing avatar that provides responses in Auslan (Australian Sign Language), enabling effective two-way communication. Our live demonstration includes gesture recognition and sign production.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility technologies; Gestural input**; • **Hardware** → **Emerging interfaces**.

KEYWORDS

gesture recognition, signing avatar, virtual assistant, smart home, Auslan, sign language, Deaf, accessibility

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1 INTRODUCTION

Voice-activated natural language interfaces are not accessible to Deaf people whose preferred language is a sign language. The Auslan Communication Technology Pipeline project (Auslan CTP) aims to create a natural language smart home for Deaf users. We have developed a prototype that: recognises emblematic gestures inspired by Auslan, via a machine learning model; triggers actions in a virtual smart home; and delivers responsive signed feedback via an interactive avatar. Our work is novel in its use of Auslan, co-design with the Deaf community, and avatar response system.

2 SYSTEM DESIGN

Auslan CTP consists of 3 modules, as shown in Figure 2. The *Recognition* module processes video input from a web camera, detects the signer's pose and performs gesture recognition (GR). The *Processing* module takes output from GR, and processes it to recognise individual commands, connecting them with specific home automation tasks. The *Production* module utilises an animated 3D human avatar that draws on a library of prerecorded Auslan signs.

In future, we aim to expand our prototype with dynamic sign recognition and conversational capabilities comparable to commercial home assistants [Korte et al. 2020].

2.1 Design Considerations

The best way to design a device for Deaf users is to design *with* them. Our team includes one Deaf person and one CODA¹, plus a 12-person Deaf Advisory Panel (DAP) who co-design with us [Korte et al. 2022a,b; Vithanage 2022; Vithanage et al. 2023].

2.1.1 Activation. Preferences included: handwave, name sign, or touching the device. This prototype activates on a handwave.

¹“Children of Deaf Adults” often have a sign language as their first language.

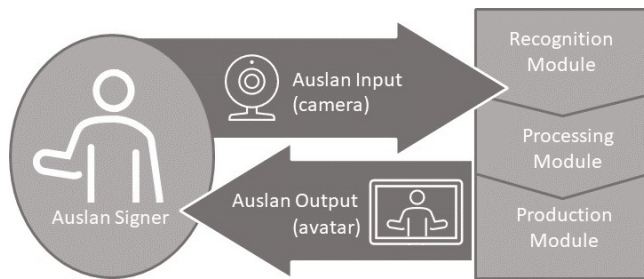


Figure 2: Auslan CTP architecture [Korte et al. 2020]

2.1.2 Avatar. Signing avatars have historically been poorly received by Deaf communities [Krausneker and Schügerl 2021; World Federation of the Deaf and World Association of Sign Language Interpreters 2018]. Avatar appearance impacts communication clarity, and potential for connection. DAP preferred a realistic avatar with clear facial expressions, and customisation options. We use photorealistic MetaHuman avatars², with customisable age, gender, skin colour, and clothing. Visibility, clarity and accuracy of hand movements are crucial for effective signing. We calibrated lighting, shadows, background colour, orientation, camera angle and distance, and clothing colour and texture, to maximise visibility.

2.1.3 Language. DAP requested dual-channel delivery of Auslan signing and toggleable English captions. Adjustable signing speed was also requested, but not yet implemented in our prototype.

2.1.4 Culture. For Zelda to be accepted by the Deaf community, it must follow Deaf cultural norms, including using sign names, signing in appropriate registers, and telling Deaf jokes [Korte et al. 2022b]. This is not yet fully addressed in our prototype.

2.2 Technical Considerations

2.2.1 Sign recognition. Sign languages use multichannel gestures, involving hand movements, facial expression and body language, with similarities between different signs, and occlusion of hands and face. Sign Language Recognition (SLR) is an active area of research [Rastgoo et al. 2021], with prior work addressing ASL SLR for home assistant devices [Santhalingam et al. 2020]. However the lack of large, appropriately labelled datasets for SLR training is a known issue [Camgoz et al. 2021; Desai et al. 2023]. Existing Auslan datasets are too small [Johnston 2008] or lack complete labeling [Shen et al. 2023]. To address this limitation, our prototype uses retrained pose estimation tools³ to recognise a limited number of emblematic gestures inspired by Auslan signs, as a proof of concept.

2.2.2 Sign variation. Auslan has “location-based, generational and community-based” variants [Korte et al. 2022b]. Individual signers may have distinct signing styles [Johnston and Schembri 2010]. SLR therefore has challenges that parallel speech recognition’s challenges with accents, dialects, and clarity [Rastgoo et al. 2021].

2.2.3 Avatar animation. All Auslan animations were recorded from native signers using full-body motion capture. Transitions between individual signs are created by interpolating between skeletal poses.

²<https://www.unrealengine.com/en-US/metahuman-creator>

³<https://google.github.io/mediapipe/>

2.2.4 Hardware. Our prototype runs on a PC or laptop and uses a web camera for input, as Deaf people prefer easily-controlled, non-intrusive SLR devices [Bragg et al. 2019; Korte et al. 2022a]. In future, we plan to deploy on a tablet; although we are limited by the requirements of the high-fidelity avatar and SLR model.

3 LIVE DEMONSTRATION

Guests will learn simple Auslan signs to control a virtual smart home⁴ and physical smart devices, with signed and captioned responses from Zelda.

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⁴Created with assets from <https://www.unrealengine.com/marketplace/en-US/product/isometric-interiors>